

Distributed LLaMA Inference System

Pratik Sarkar

Arkadip Kansabanik

Rik Halder

M.Sc Data Science & Artificial Intelligence

April 20, 2026

Objective

Build a distributed system where one machine acts as **Root/Master Node** and multiple systems act as **Worker Nodes** to run LLaMA models faster using CPU parallelism.

- Local offline LLM deployment
- Multi-node distributed inference
- Lower memory load per machine
- Faster token generation
- Scalable architecture

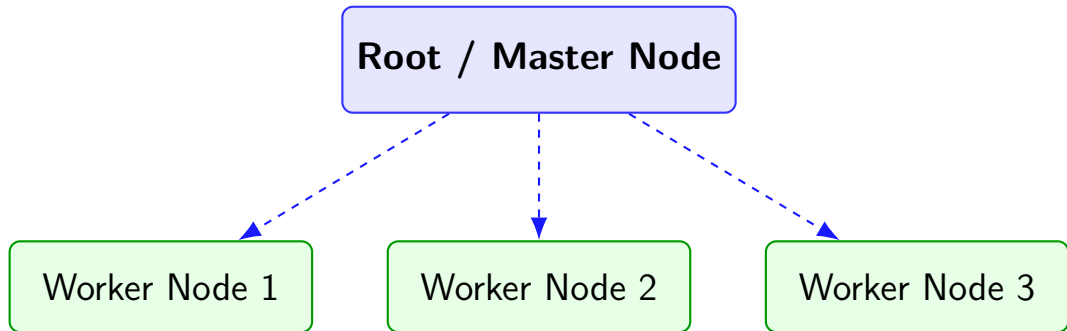
Problem Statement

- Large Language Models require high RAM and compute resources.
- Running bigger models on a single PC is difficult.
- GPU infrastructure is expensive.
- Need a low-cost distributed local AI system.

Solution

Split inference workload across multiple connected machines over LAN.

System Architecture



Master coordinates requests, workers process model layers.

Execution Workflow

- ① Start worker nodes on each machine.
- ② Root node connects all workers.
- ③ Network synchronization initialized.
- ④ Model weights loaded across nodes.
- ⑤ User sends prompt via local API.
- ⑥ Workers collaboratively compute inference.
- ⑦ Final response returned.

Real Deployment Results

Observed Logs

- Listening on 0.0.0.0:9998
- Root node connected successfully
- Network initialized
- Weights loaded
- CPU detected: AVX2

LAN Connectivity

Ping latency observed near **0.1 ms** between nodes.

Research Benchmark Results

Devices	1 Node	2 Nodes	4 Nodes
LLaMA 2 7B Total Time	1312 ms	793 ms	494 ms
Inference Time	1307 ms	739 ms	458 ms

Insight

Increasing worker nodes significantly reduces inference latency.

Advantages

- Lower RAM usage per node
- Faster generation
- Offline deployment
- Cost effective
- Easily scalable
- CPU cluster support
- Better utilization
- Research ready

Challenges Faced

- Port connection issues
- No route to host errors
- Worker synchronization delay
- Model path mismatch
- Thread optimization
- Network interruptions

Industry-Level Enhancements

- Dockerized deployment
- Kubernetes autoscaling
- Monitoring dashboard
- Authentication API gateway
- Load balancing
- RAG integration
- AI Log Analyzer chatbot

Applications

- Enterprise AI Server
- Chatbot Hosting
- Offline AI Labs
- Secure On-prem LLM
- Edge AI Devices
- Research Clusters
- Multi-user Inference
- Education Labs

Next Version

- GPU + CPU Hybrid Cluster
- Automatic Node Discovery
- Multi-model Support
- High-speed Interconnect
- Production Scale Deployment

Thank You